Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces
  Reading • 18 min
- Linear Independence, Basis, and Rank
  Ungraded App Item • 20 min
- Linear Independence
  Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings
  Reading • 1h
- Linear Mapping: Part 1
  Video • 8 min
- Linear Mapping: Part 2
  Video • 8 min
- **Linear Mapping**
  Reading • 5 min
- Matrix Representation
  Reading • 6 min
- Practice Quiz: Linear Mapping
  Graded Assignment • 18 min

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors
  Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications
  Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition
  Reading • 5 min
- Solved Examples
  Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD)
  Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors
  Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

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Linear Mapping

A linear mapping (or linear transformation) Φ:V→W is a function that takes vectors from one vector space and maps them to vectors in another vector space while preserving the operations of vector addition and scalar multiplication. This means that the way vectors are combined and scaled in the original space is maintained after the transformation.

Key Properties

Preservation of Addition:

- If you have two vectors x and y in a vector space V, a linear mapping Φ:V→W will satisfy:

$$\Phi(x+y) = \Phi(x) + \Phi(y)$$

- This property ensures that the result of adding two vectors and then applying the transformation is the same as applying the transformation to each vector separately and then adding the results.

Preservation of Scalar Multiplication:

- For any scalar and vector x in V, a linear mapping Φ:V→W will satisfy:

$$\Phi(\lambda x) = \lambda\Phi(x)$$

- This property ensures that scaling a vector before applying the transformation gives the same result as applying the transformation and then scaling the transformed vector.

Example of Linear Mapping

Consider a simple example in $\mathbb{R}^2$:

Let Φ: $\mathbb{R}^2 \to \mathbb{R}^2$ be defined by:

$$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 3x_2 \end{pmatrix}$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

and

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Check the properties:

For addition:

$$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \Phi\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 3x_2 \end{pmatrix} + \begin{pmatrix} 2y_1 \\ 3y_2 \end{pmatrix} = \begin{pmatrix} 2\left(x_1 + y_1\right) \\ 3\left(x_2 + y_2\right) \end{pmatrix} = \Phi\begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}$$

For scalar multiplication:

$$\Phi\left(\lambda\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \Phi\begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix} = \begin{pmatrix} 2\lambda x_1 \\ 3\lambda x_2 \end{pmatrix} = \lambda\begin{pmatrix} 2x_1 \\ 3x_2 \end{pmatrix} = \lambda\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

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